

Analysis of the Determinants of Economic Output and Development

From 2000-2021

Introduction

Gross Domestic Product (GDP) per capita is the most widely used measurement of a country's economic status, representing the final economic value from all of the goods and services within a country's borders in a year per person. Meanwhile, the Human Development Index (HDI) is the most widely used indicator of a country's quality of life, incorporating elements of health, education, and wealth. Although the two measures are directly related to factors like health, education, and income, there are many characteristics that influence both GDP per capita and HDI over time. Determining which factors are most closely related to those measures, and how much they affect the variables is crucial to give a more thorough view of the determinants of wealth and quality of life.

The purpose of this project was to identify the variables most closely related to GDP per capita and HDI and to compare the importance of those variables using both econometric methods, and machine learning models. Ordinary least squares (OLS) regressions and linear regressions were used to examine the relationships between the selected explanatory variables and the two dependent variables. Subsequently, gradient boosting was used to evaluate nonlinear patterns, while feature importance and SHAP were used to determine the impact of each independent variable on the outcomes. By using these approaches, this project not only gives an econometric background with interpretable coefficients, but also a fundamentally different look at what factors play the largest roles in shaping economic output and development status.

Data

When it came to selecting variables to examine their effects on GDP per capita and HDI, I implemented a few criteria for determining whether I'd add a variable. Firstly, the variable needed to be able to feasibly impact the dependent variable in some way. Secondly, it couldn't be directly involved in the construction of the dependent variable (this is particularly relevant for HDI, as it disqualified GDP, as it uses GNI per capita, along with health metrics like life expectancy, and educational attainment metrics like years of schooling). Thirdly, the variable couldn't be a result of changes in the dependent variable, as this would be an example of reverse causality. Lastly, the variable couldn't be a "bad control," which in econometrics refers to a variable which is the causal linkage between an independent variable, and a dependent variable. For instance, if you try to examine the effect of smoking on lung cancer, including a metric for lung health is a bad control because smoking causes lung cancer by making your lung health worse.

These criteria ultimately led to me excluding quite a few variables. Firstly, for the GDP model, I removed domestic credit to the private sector (which was the chosen financial variable I worked with for some time) because increased financialization is just as much a result of economic development as it is a cause. Additionally, I removed dollar output per hour (a measure of labor productivity), because it qualifies as a bad control, being downstream variables like education and GDP, because increased education leads to increased productivity, which leads to a higher per capita GDP. As a result, those variables (along with GDP per capita) were automatically removed from the HDI model because HDI is a holistic metric which uses Gross

National Income per capita (PPP), which is almost identical to GDP per capita. On top of that, I removed average years of schooling, life expectancy, and child mortality (under 5 years old) because they're all major factors in computing HDI. When it came to GDP, life expectancy and infant mortality were removed from the model as well because they're also examples of reverse causality. This is because increased life expectancy doesn't make a country rich, a country becoming rich leads to a better healthcare system which leads to a higher life expectancy (and the same applies with child mortality).

The final datasets included GDP per capita at purchasing power parity and HDI as the dependent variables^{[1][2]}. The independent variables were sorted into a few categories. The first category was comprised of demographic variables, which include a country's dependency ratio (the number of people under 15 or over 64 divided by the population between 15-64), population growth rate, employment/population ratio (the percentage of people over 15 who work), the urbanization rate, and the percent of people working in agriculture^{[3][4][5][6]}. The model also included average years of schooling as a human capital variable (only for the GDP model), as well as variables for economic structure like trade, fixed capital formation, and resource rents (all expressed as a percentage of GDP), along with per capita energy consumption (in kilowatt hours per person)^{[7][8][9][10][11]}. I included the rule of law index as an institutional variable, which is crucial as a reflection of trust in an economic system, along with the global commodity price index (which I expressed as a percent change year over year) in order to see how the resource intensity of an economy affects its performance when resource prices are increasing^{[12][13]}. Lastly, I made a few interaction terms, which multiply one variable by another to give information on how the relationship between an independent and dependent variable changes when that independent variable is contextualized. One example was with population growth, which was

multiplied by the dependency ratio so that the effect of population growth was more thoroughly understood based on whether the increased population was a result of a country having more adults who could work, or more children who are a fiscal challenge for a country to navigate.

Methods

The main methods used throughout the project were an ordinary least squares (OLS) regression, a linear regression with an 80/20 train-test split, a gradient boosting model, and SHAP and PDP based model interpretations. The regression was used to estimate and quantify the relationships between the independent variables, and the two dependent variables, GDP per capita, and HDI. The linear regression model with the train-test split was then made and its findings were compared to the OLS regression. Meanwhile the other machine learning models were used to find nonlinear relationships and the variables which had the most importance in terms of influencing the independent variables

Linear Regression

A linear regression refers to a model that tries to identify the relationships between a dependent variable and an independent variable(s). It determines the relationship between the two by constructing a linear function (with the slope representing the effect of an independent variable) based on minimizing the sum of squared errors between the actual and predicted values of Y ^[14]. A linear regression may predict someone's income based on education attainment, parental income, median income in their city, and other factors. In a multivariate regression (like the ones used in the project), the coefficients refer to the effect that a change in the independent

variable has on the dependent variable, when all other variables are held constant, which gives it the ability to simultaneously quantify the effects of multiple independent variables. In the case of this project, the same underlying linear regression model was used for both the OLS regression, and the linear regression on scikit-learn.

In order to perform the regressions, I took each independent variable and either plotted them against the dependent variable, or compared the RMSE (root mean squared error, which is the square root of the averaged differences between the fitted values and the actual values) of the linear specification, to the squared, cubic, and natural logarithmic specifications^[15]. When the alternative specifications didn't show a substantial improvement in comparison to a linear specification, the linear form was used. In the case of both the GDP per capita and HDI models, there were a good amount of transformations which needed to be done to linearize the variables for a regression. Those transformations were used for both the OLS model, and the scikit-learn regressions.

To perform the regressions in scikit-learn, the observations were divided into a training portion, and a testing portion. For the training portion, 80% of the observations were used for training the model to find relationships, while 20% were used for testing to predict the dependent variable. The models were compared to the OLS model using R^2 and compared to each other using R^2 and RMSE. Additionally, I found the coefficients for both the training and testing model, and compared their values with that of the original OLS model which showed whether the relationships held up in the two subsamples. The purpose of splitting the data into training and testing categories is to make a model that helps learn from the data, and identifies patterns and relationships, while the latter category is used to see how well the model performs when generalizing to unseen data^[16]. The OLS model wasn't technically necessary for the project,

however, I made it to cross reference the two ML models against a baseline containing all of the data.

Gradient Boosting and Feature Importance

After making the linear regressions using scikit-learn, I moved on to gradient boosting for the two models in order to find relationships which can't easily be identified with linear regressions. Gradient boosting is an algorithm that uses ensemble learning to sequentially combine many decision trees to create a single model^[17]. Ensemble learning combines multiple models in order to improve predictive performance by training based on prior mistakes, giving more weight to misclassified models by identifying issues and optimizing to correct for those mistakes. Gradient boosting takes weak decision trees and sequentially goes through them to make a more accurate model with fewer errors. Since gradient boosting combines many different decision trees, it can more effectively capture relationships. However, since this can result in overfitting (which happens when a model functionally memorizes training data, but can't identify patterns, thus failing to adapt to new data), it requires hyperparameter tuning, where you plug in different values for factors like the maximum depth of a decision tree, in order to make a model which is optimized in terms of many factors. The hyperparameters I evaluated for this project include the learning rate, maximum depth of the decision trees, minimum sample leaf and split, number of estimators, and the subsample proportion.

The learning rate refers to the rate at which the model learns from new observations^[18]. If a model has a very low learning rate, it may see a set of new observations, and instead of using them to help change its impression of the underlying patterns in the data, it might just find them to be an anomaly, while a very high learning rate will see new observations that differ from the

rest of the data, and overcorrect, fundamentally reshaping its impression of the data's patterns. The maximum depth refers to the maximum number of levels in each decision tree following from each node, which is the single point at the top of a decision tree^[19]. A maximum depth of 3 means that there would be no more than 3 levels below the node in each decision tree. The minimum sample leaf is the minimum number of samples that need to be at a node. It's done to ensure that there aren't nodes with very few samples in them, which can lead to overfitting. The minimum sample split is the minimum number of samples needed to split a node. For instance, if a node has 5 samples and the minimum sample split is 8, then the node terminates there and no more splits are done, but if it has 14 samples, then it'll split, without it having to be a 50/50 split (such as with an 8-6 split, in which case there would be one more split). Lower minimum sample split values can lead to overanalyzing small samples and overfitting, so it's important to optimize for it. The number of estimators refers to the number of total decision trees throughout the entirety of all of the forests, with more trees typically leading to models that have better predictions, but take a longer time. Lastly, the subsample is essentially the optimal proportion of the sample used for each training round during the model.

Hyperparameter tuning was mainly done using randomized searches. This randomly samples combinations of hyperparameter values, and evaluates them using cross validations on the training data (I used 3 cross validations). This allows for many combinations to be examined without looking at every single possible combination. The models were then compared with R^2 as the method of comparison. After finding the final hyperparameters, the models were evaluated using R^2 , RMSE, mean absolute error, and mean squared error, which were used for the training data as well. Once that happened, the training and testing performances were compared to see if the model was overfitting based on the training data. I exclusively used grid searches during the

hyperparameter tuning portion of the SHAP process, albeit, in that context, they takes considerably less time to work than during the gradient boosting portion, so it didn't function as the main bottleneck for the project, even compared to the random search used in the gradient boosting section.

SHAP and Partial Dependence Plots

Subsequently, the models were used to test for feature importance. Feature importance is essentially a score given for each independent variable in the model to show how important they are in influencing the dependent variable^[20]. These values tell you which variables give you the most information about the dependent variable (which is very important from a decision making perspective), and they can tell you what to remove from the model to save time without sacrificing much in terms of quality, but they don't tell you anything about the magnitude or direction of the effect, nor do they tell you the causal impact of the variable.

I then used SHAP (SHapley Additive exPlanations) to give observation level numerical interpretations for each of the variables in the model. SHAP is a game theory tool used to tell you the effect each independent variable has on the dependent variable^[21]. It does this by distributing credit for model outputs among variables by determining their effects on the dependent variable when a variable "joins" a model. A variable "joins" the model when the value of its feature is revealed, while it hasn't joined when that value is left unknown. Similarly to feature importance, it differentiates between coefficient magnitude and importance in a linear model. However, unlike feature importance and linear regression models, it tells you the precise effect that each independent variable has in a given observation, which isn't based on a constant slope or coefficient value. These values were then used to generate waterfall plots, which show

you how the discrepancies between an observation's value for an independent variable and the sample average ultimately affect the final value of the dependent variable by way of showing the difference between the baseline value (where no variables have "joined" a model) and the model's estimate for the value.

Afterwards, for the GDP and HDI models, I calculated the mean absolute SHAP values for each variable to determine their average contribution to the dependent variable. The three variables with the largest SHAP values on average were used to make partial dependence plots (PDPs). PDPs show you the average predicted value for the dependent variable when one of the independent variables changes across its range, which shows you the model's estimate of the relationship between a feature and the dependent variable while holding the values of other variables constant.

Visualizations

After completing the statistical analysis and machine learning models, I used Tableau to create two dashboards for GDP per capita and HDI. Both dashboards compared the value of the dependent variable for the first available year for each country to the last available year.

Additionally, they both had treemap plots for the feature importance of each variable, as well as a circle plot for the SHAP mean absolute values for each variable, as well as a sample waterfall plot obtained from python.

Results

GDP Per Capita at Purchasing Power Parity (PPP)

Regression Models

The GDP per capita model found a few variables that exhibited clearly positive relationships with economic output, including average years of schooling, rule of law, per capita energy consumption, and the share of the population living in urban areas. Energy consumption (which was expressed as a logarithmic term) had the strongest effect on GDP. Some variables exhibited diminishing returns like fixed capital formation (which becomes a drag on GDP once it reaches 21% of a country's GDP) and employment population ratio. This may mean that too much money into capital can take away from other areas of an economy, including consumption. Additionally, it shows that each unit of fixed capital (and labor) adds less value to economic output, which aligns with typical Cobb-Douglas models of the effect of capital and labor on output, where they discuss the decreasing marginal product of capital and labor^[22].

The dependency ratio also exhibited a negative relationship, and the effect gets worse the faster the population grows (due to its interaction term). Population growth itself has a very complex relationship with GDP, itself depending on the country's dependency ratio and other factors. Resource rents didn't have much of an effect on GDP, including with its interaction term involving the change in the global commodity price index. Trade intensity actually had a statistically significant negative effect on GDP, but the effect wasn't strong, with even large increases in trade having essentially no isolated effect on GDP. The data also showed that fewer workers in agriculture has a very positive effect on GDP.

The scikit-learn regression models returned similar values for model performance across the training and testing datasets, compared to the full dataset, based on their R^2 values. However, the coefficients varied when it came to model performance. When it came to the training dataset, most coefficients were within roughly 10% of the OLS values, with the main exception being from dependency ratio. When it came to the testing dataset, just under half of the variables had coefficients that were very similar to the OLS values, while many differed by more than 50%.

Gradient Boosting and Feature Importance

Hyperparameter tuning resulted in a final gradient boosting model with 12,000 estimators (or decision trees), a subsample of 0.4, a minimum sample split of 2, a minimum sample leaf of 5, a maximum depth of 5 levels, and a learning rate of 0.0075. The gradient boosting model had a far higher R^2 than in the regression models (roughly 0.905), likely because it was better at searching for nonlinear relationships. While the training R^2 model had an R^2 of essentially 1 (0.99996), the test R^2 model had an R^2 of 0.9925. This may be because of data leakage, which may've resulted from the random sampling giving data from each year instead of just giving all of the data for 80% of countries^[23]. As a result, I made a group shuffle split for the GDP and HDI models, which groups data into one set or another based on the country^[24]. In the group shuffle split, the model still had an R^2 of 0.9457.

Feature importance values showed that per capita energy consumption was the most important variable when it came to GDP per capita, with a value of 0.6998, far higher than any other variable. The percentage of people working in agriculture was the next most important variable, with a value of 0.1549, then urbanization at 0.0269, then the employment population ratio at 0.0237. Afterwards, every other variable had a value of less than 0.02.

SHAP and Partial Dependence Plots

SHAP analysis was used to find the impact of the variables on the GDP model. The model for the SHAP analysis found that the best hyperparameters were 28,000 estimators, a subsample of 0.7, a learning rate of 0.075, a maximum depth of 5, and a column sample of 0.8 (meaning that 80% of the columns were used for each observation to help train the model by making it adapt to unseen values). The SHAP values were then used to give waterfall plots to show how each variable moves the predicted value of GDP away from the baseline value (expressed as a log because GDP is best as a logged variable), with an example shown below:

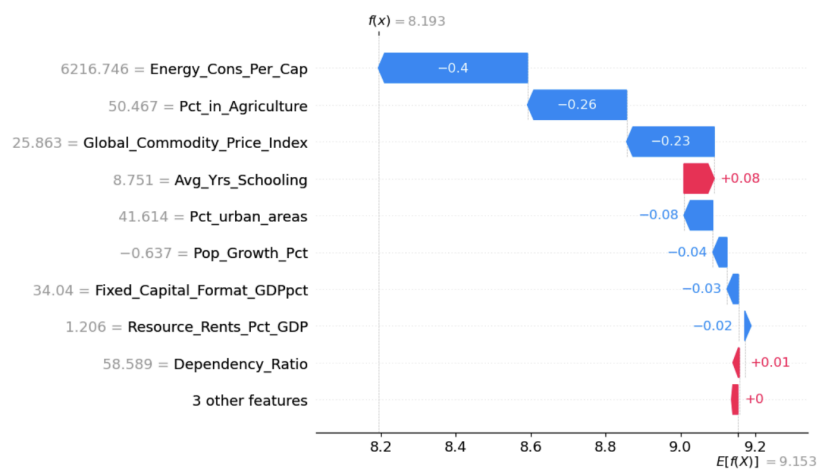


Figure 1: SHAP Waterfall Plot For A Country-Year Combination (Albania 2000)

In the case shown above, the actual prediction for log(GDP per capita) is 8.193 (\$3,656 per capita), while $E[f(x)]$ (the baseline estimate when none of the values for the variables are shown) is 9.153 (\$9,443 per capita). In that plot, energy consumption had a negative contribution to the SHAP value because it's far below the sample average. In subsequent country-year values

for Albania, despite energy consumption being below the sample average, it actually had a positive effect. This may've been because it learned the country-year patterns for Albania and recognized that small increases in energy consumption meant its per capita GDP was increasing, or it was a country specific phenomenon. I looked at a few different country-year combinations, and they all showed that a below average energy consumption value corresponded to a negative effect on GDP per capita.

Once a couple waterfall plots were shown, I looked at the mean of the SHAP values (as an absolute value so the effect doesn't dilute) for each of the variables in order to see which ones had the largest effect on GDP per capita. Once again, per capita energy consumption had easily the largest effect at $|0.66|$ on average, followed by the percent working in agriculture (at $|0.2|$), as well as urbanization, and the change in the global commodity price index at $|0.08|$ for both. After finding how large the SHAP values of each variable was on average, I made Partial dependence plots for the 3 variables with the largest total effects with an example shown below:

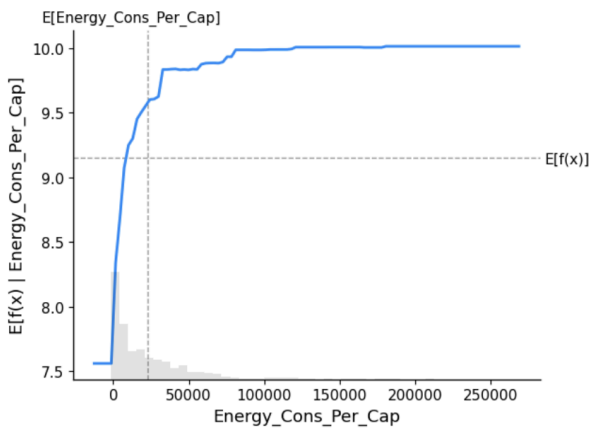


Figure 2: Partial Dependence Plot For Per Capita Energy Consumption

The plot above shows the average predicted value for the log of GDP per capita at different levels of energy consumption. For instance, when it reaches the expected value of per capita energy consumption (roughly 25,000 kWh per capita), the average predicted log value of GDP is roughly 9.6 (\$14,765). Similar plots were made for the percent of people working in agriculture, and the urbanization rate.

Human Development Index

Regression Models

The HDI model gave several relationships that were very similar to those in the GDP per capita model. Per capita energy consumption, urbanization, and improved rule of law have clear positive effects on HDI. Trade intensity also had a negative but small relationship with HDI. Population growth also had a very complex effect on HDI, while fixed capital formation and the employment population ratio exhibited diminishing returns, as was the case with the GDP model. Meanwhile a higher share of employment in agriculture had a strongly negative effect on HDI, as did a higher dependency ratio, and increased resource intensity. Unlike the GDP model, education wasn't included as a predictor because education directly feeds into HDI calculations.

The train-test models in scikit-learn returned very similar values in terms of the model strength, as the 3 models had similar R^2 values. The coefficients estimated from the training dataset were very similar to those from the full OLS model, also typically being within 10% of the values from the OLS model. Meanwhile, the coefficients used from the testing dataset were generally within 30% of the full OLS model, considerably more consistent than in the GDP model.

Gradient Boosting and Feature Importance

When it came to hyperparameter tuning, the final gradient boosting model used 16,000 estimators, had a subsample of 0.5, a minimum sample split of 6, a minimum sample leaf of 5, a learning rate of 0.0075, and a maximum depth of 10. Similarly to the GDP model, the R^2 was far higher than in the regression models, which was also likely a result of both looking for nonlinear relationships, and having 80% of the data from all of the countries, as opposed to having 100% of the data from 80% of countries. Once again, both the training and testing models had R^2 values above 0.99, likely indicating data leakage. Similarly to the GDP model, I performed a group shuffle split model, which showed an R^2 of 0.9049, well below the original values from the gradient boosting models, but still indicative of the fact that the models do a good job of properly estimating the model.

Feature importance results once again found per capita energy consumption to be easily the most important factor influencing the dependent variable, with a value of 0.7163, followed by the percent in agriculture at 0.1402, the dependency ratio at 0.0565, and the rule of law index at 0.0296, with the remaining variables having feature importance values of less than 0.02.

SHAP and Partial Dependence Plots

The final SHAP values for the GDP model contained 65,000 estimators, a subsample of 0.8, a learning rate of 0.075, a maximum depth of 5, and a column sample of 1.0. Afterwards, the SHAP values were used to give waterfall plots, which show how each variable moves the predicted value of HDI away from the baseline value of HDI, with an example shown below:

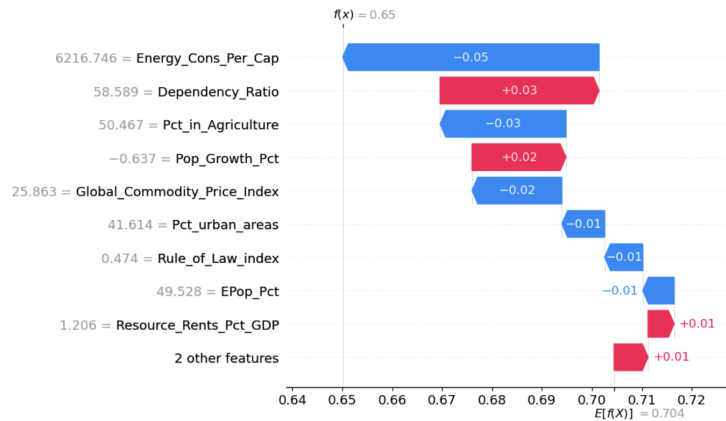


Figure 3: SHAP Waterfall Plot For A Country-Year Combination (Albania 2000)

In the case shown above (Albania 2000), the actual prediction for HDI is 0.65, while $E[f(x)]$ is 0.704. The SHAP contribution of energy consumption is also negative because it's far below the sample average. In subsequent country-year values for Albania, despite being below the sample average, it actually had a positive effect, which may've been for similar reasons as in the GDP model, like it learning the pattern of the variables for Albania.

Once the waterfall plots were finished, the mean absolute SHAP values were found, and they identified per capita energy consumption as easily the most important variable, with an effect of $|0.09|$ on average, followed by the dependency ratio at $|0.03|$, and the percent in agriculture at $|0.02|$. After finding how large the SHAP values were on average, I made partial dependence plots for the 3 variables with the largest effects, as with the GDP model, with the example for dependency ratio shown below:

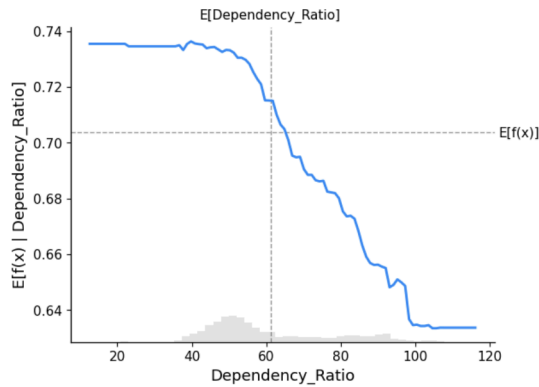


Figure 4: Partial Dependence Plot For Dependency Ratio

The plot above shows the average predicted value for HDI when the dependency ratio is at a certain level. For instance, at a dependency ratio of roughly 61% (which is the expected value), the average value of HDI is roughly 0.715. Similar plots were made for the percent of people in agriculture, and the urbanization rate.

Conclusion

Regression Models

The findings from the baseline OLS model offer several possible suggestions for facilitating both an increased economic output, and economic development, as the policy implications from both the GDP and HDI models are very similar. The main focus is to get more people into high quality education and urban areas, as it presents a great opportunity for people to both develop skills, and be able to access better jobs. While agricultural jobs add little value to the economy, and don't present many opportunities for human capital development and upward

mobility, moving to urban areas puts people in closer proximity to superior educational systems, and offers a wider variety of high quality jobs. That said, urban areas carry their own risks like increased disease burdens, infrastructure pressures, and elevated carbon emissions from high energy consumption. Therefore, the main focus needs to be to increase investment into schooling and housing construction, as well as improving public health systems in order to give people ample opportunities to access high skilled employment, and to ensure that the downsides of moving to urban areas are mitigated.

In both models, increasing capital investment helps to a point, and improving the rule of law in a country has its own positive effects. In the case of the GDP model, this is likely a result of it fostering more trust in institutions (economic and otherwise), which is very important for the stability of a capitalist economy. Meanwhile, in the case of the HDI model, it likely results from the aforementioned factors, as well as improved public administration, law enforcement, and anticorruption mechanisms, which can improve quality of life.

Lastly, energy consumption had the most significant effect on economic output, and it needs to be in abundant supply, however, since global temperatures are rising in spite of so many countries not being industrialized, increased economic development can come with significant environmental and ecological challenges, so it's important to concurrently develop low carbon energy sources and electrical grids to balance the effects of increased energy demand.

The training and testing models in scikit-learn returned performed relatively similarly to the OLS model in both the GDP and HDI models. When it came to the training models, they returned broadly similar coefficient values, while in the testing models, they performed somewhat differently. The GDP testing model had many variables whose coefficients differed from the OLS model by more than 50% which may've been a result of the fact that it was a small

subset of the overall dataset, as the model didn't suffer from a lack of external validity, because it still generalized well to the unseen data, but was likely handling different data. Meanwhile, the coefficients in the HDI testing model were typically within about 30% of the coefficients from the OLS model. This means that the random data used for the testing portion was likely more similar to the full dataset than it was for the GDP model.

Gradient Boosting, Feature Importance, and SHAP

In the gradient boosting model, the R^2 values were far higher than in the regression models. This is to be expected to a degree, because the gradient boosting model takes a more thorough look at all of the data to determine which relationships best suit the model. However, the R^2 values were well above 0.99 for both the training and testing models, which was likely a result of the fact that the models had information on all of the countries for most of their years, and could therefore get a solid grasp of the relationships between the independent and dependent variables for every country. However, this may mean that the model essentially learns the relationships for each country specifically, instead of understanding the broader relationship between the features and the outcomes. The group shuffle split was used to correct for this, with the country being used as the grouping variable. As a result, the R^2 for both of the models decreased (down to roughly 0.945 and 0.905), however, the models still did a very good job predicting the relationships between variables, while being able to generalize the data to unseen countries at large.

In the case of feature importance, both models find energy consumption and the population share in agriculture to be by far the most important variables, giving substantially more information than any other variables, especially as it pertains to energy consumptions. The

models also find urbanization and the dependency ratio to also be quite helpful, and the emphasis which they place on the former two variables is larger with respect to feature importance than is the case with the regression models. When it comes to SHAP models, they offered broadly similar conclusions, as energy consumption was the most important variable by a wide margin, while agricultural employment was still very important (especially for the GDP model), along with urbanization, and the dependency ratio (which was the 2nd most important variable for the HDI model). The main discrepancy was that the change in the global commodity price index was very important to the GDP model, which is more in line with what I would've predicted as low income countries with relatively unsophisticated economies heavily rely on the global price of resources, which is part of what keeps most of them from becoming wealthy. However, the main takeaway from the feature importance and SHAP values is just how important energy consumption is, with the partial dependence plots showing this as well. In the partial dependence plots, there is a very wide range of values for the dependent variable when given the value of per capita energy consumption, far more than with the other most important variables, because of just how much information it gives the models.

Limitations

Although I think the choice of variables in the model was good, I think that many of the models suffered because I predominantly used functions which randomized the data in its entirety, rather than randomizing by country or by year. Although the group shuffle split did group the train-test split by countries (so all of the data from a country like the US was in one section), it was generally a minor portion of the project. I would've liked to see how the

relationships changed over time, because per capita energy consumption in some wealthy countries has actually plateaued or decreased with time. Those patterns can certainly have an effect on the relationships, as some variables become more or less important with time. This is especially important as global averages for the variables increase and values converge, which means that the variables provide you with less information about why development outcomes differ. I think a very interesting extension would be to examine how the relationships differ based on a country's development status. For instance, separate regressions could be made for countries with GDP per capita (PPP) levels below \$10,000, between \$10,000-\$25,000, between \$25,000-\$50,000, and above \$50,000 (and the same could be done with HDI). This would give you the opportunity to see how relationships between variables change at different levels of development. Another approach could be to examine how the effects of variables change over time by creating models for 2000-2005, 2005-2010, 2010-2015, and 2015-2021. As a result, this could show you which of these determinants were more important, and which ones are more important for today, and can improve the quality of policy recommendations.

However, conducting these analyses would have substantial computational burdens. Even the models I made were incredibly time consuming to run, especially for gradient boosting, which is crucial for offering policy implications. Nonetheless, these approaches are all very helpful extensions of the project because they can offer a better understanding of the most important factors influencing development, based on both the development status of the country, and the point in time in which the data is analyzed.

References

- [1]: *GDP per capita, PPP (current international \$)*. World Bank Open Data. (n.d.-a).
<https://data.worldbank.org/indicator/NY.GDP.PCAP.PP.CD>
- [2]: *Human development index*. Our World in Data. (n.d.-b).
<https://ourworldindata.org/grapher/human-development-index>
- [3]: *Total dependency ratio*. Our World in Data. (n.d.-b).
<https://ourworldindata.org/grapher/age-dependency-ratio-of-working-age-population>
- [4]: *Population growth (annual %)*. World Bank Open Data. (n.d.).
<https://data.worldbank.org/indicator/SP.POP.GROW>
- [5]: *Employment rate*. Our World in Data. (n.d.-a).
<https://ourworldindata.org/grapher/employment-to-population-ratio>
- [6]: *Share of population living in urban and rural areas*. Our World in Data. (n.d.-c).
<https://ourworldindata.org/grapher/share-urban-and-rural-population>
- [7]: *Years of schooling - by level of Education and Gender*. Our World in Data. (n.d.-c).
<https://ourworldindata.org/grapher/years-of-schooling>
- [8]: *Trade as a share of GDP*. Our World in Data. (n.d.-e).
<https://ourworldindata.org/grapher/trade-as-share-of-gdp>
- [9]: *Gross fixed capital formation (% of GDP)*. World Bank Open Data. (n.d.).
<https://data.worldbank.org/indicator/NE.GDI.FTOT.ZS>
- [10]: *Total natural resource rents*. Our World in Data. (n.d.-b).
<https://ourworldindata.org/grapher/natural-resource-rents>
- [11]: *Energy mix*. Our World in Data. (n.d.).
<https://ourworldindata.org/grapher/per-capita-energy-use>

[12]: *Rule of law index*. Our World in Data. (n.d.).

<https://ourworldindata.org/grapher/rule-of-law-index>

[13]: *Global price index of all commodities*. FRED. (2026, July 13).

<https://fred.stlouisfed.org/series/PALLFNFINDEXQ#>

[14]: *Linear regression in machine learning*. GeeksforGeeks. (2026, June 4).

<https://www.geeksforgeeks.org/machine-learning/ml-linear-regression/>

[15]: *Multivariate statistical details*. IBM. (2025, January 31).

<https://www.ibm.com/docs/en/planning-analytics/2.0.0?topic=explained-multivariate-statistical-details>

[16]: Sivakumar, M., Parthasarathy, S., & Padmapriya, T. (2024). Trade-off between training and testing ratio in machine learning for medical image processing. *PeerJ Computer Science*, 10. <https://doi.org/10.7717/peerj-cs.2245>

[17]: Clark, B., & Lee, F. (2025, November 17). *What is gradient boosting?*. IBM.

<https://www.ibm.com/think/topics/gradient-boosting>

[18]: Belcic, I., & Stryker, C. (2025, November 17). *What is learning rate in machine learning?*.

IBM. <https://www.ibm.com/think/topics/learning-rate>

[19]: *Hyperparameters of random forest classifier*. GeeksforGeeks. (2025, July 3).

<https://www.geeksforgeeks.org/machine-learning/hyperparameters-of-random-forest-classifier/>

[20]: *Understanding feature importance in machine learning*. Built In. (n.d.).

<https://builtin.com/data-science/feature-importance>

[21]: An introduction to explainable AI with shapley values. (n.d.).

https://shap.readthedocs.io/en/latest/example_notebooks/overviews/An%20introduction%20to%20explainable%20AI%20with%20Shapley%20values.html

[22]: Science Direct. (n.d.-b). Cobb-Douglas Production Function - An Overview | Sciencedirect topics.

<https://www.sciencedirect.com/topics/engineering/cobb-douglas-production-function>

[23]: Mucci, T. (2025, November 17). *What is data leakage in machine learning?*. IBM.

<https://www.ibm.com/think/topics/data-leakage-machine-learning>

[24]: *How to generate a train-test-split based on a group Id?*. GeeksforGeeks. (2025a, July 23).

<https://www.geeksforgeeks.org/machine-learning/how-to-generate-a-train-test-split-based-on-a-group-id/>